COSC 39: Theory of Computation

Credit Statement: Talked to Claude.

Problem 1. Let G be an (either undirected or directed) graph. A *Hamiltonian path* in G is a path that visits every single vertex in G exactly once. Consider the following problems:

DIRECTEDHAMPATH

- **Input:** A directed graph G
- **Output:** Is there a Hamiltonian path in G?

HAMPATH

- **Input:** An undirected graph G
- **Output:** Is there a Hamiltonian path in G?

1. Describe a polynomial-time reduction from HAMPATH to DIRECTEDHAMPATH.
2. Describe a polynomial-time reduction from DIRECTEDHAMPATH to HAMPATH.

   [Hint: After trying the first reduction that comes to mind, check the no-to-no direction first: Can you recover from a Hamiltonian path on the constructed undirected graph a corresponding Hamiltonian path on the input directed graph?]

Solution.

1. Given an undirected graph $G = (V, E)$, we construct a directed graph $H = (V, E')$ as follows,
   - Every vertex in G is a vertex in H,
   - For an edge in G between vertices v, w we create two directed edges $v \to w$ and $w \to v$ in H.

($\implies$)

If G has a Hamiltonian path $p = v_1, v_2, \ldots, v_n$ then the corresponding vertices $v_1, v_2, \ldots, v_n$ in H also have a directed path $v_1 \to v_2 \to \cdots \to v_n$.

($\impliedby$)

If H has a Hamiltonian path $p = v_1 \to v_2, \to \cdots \to v_n$ then we know that each directed edge (v_1, v_2) came from an undirected edge between v_1 and v_2 in G, thus there is a Hamiltonian path $v_1, v_2, \ldots, v_n$ in G.

That is, G has a Hamiltonian path if and only if H has a Hamiltonian path. (And this construction takes polynomial time because we are only adding two edges per edge.)

2. Given a directed graph $G = (V, E)$ we construct an undirected graph $H = (V', E')$ as follows,

   - For every vertex $v \in V$ we create a gadget with three vertices v_{in}, v, v_{out} in V' and with edges $\{v_{in}, v\}$ and $\{v, v_{out}\} \in E'$.
   - For every edge $v \to w$ in G we create an edge between v_{out} and w_{in}.

   ($\implies$)

   If G has a Hamiltonian path $p = v_1 \to v_2 \to \cdots \to v_n$, then H has a Hamiltonian path by construction,

   $$v_{1_{in}},\ v_1,\ v_{1_{out}},\ v_{2_{in}},\ v_2,\ v_{2_{out}},\ \ldots,\ v_{n_{in}},\ v_n,\ v_{n_{out}}$$

   ($\impliedby$)

   If H has a Hamiltonian path p then we claim that the path traverses each gadget (v_{in}, v, v_{out}) consecutively because v's only neighbors in H are v_{in} and v_{out}, so any Hamiltonian path visiting v must come in from one and leave from the other and since every vertex v must be visited the three nodes are visited consecutively.

   We also claim that all gadges are traversed in the same direction, that is if for some v, p enters at v_{in} and exits at v_{out} then it must also enter at w_{in} and exit at w_{out} for all w. This is because v_{in}'s neighbors are v and nodes of the form w_{out}, and v_{out}'s neighbors are v and nodes of the form w_{in}.

   Assume some gadget is entered at v_{in} and exited at v_{out}, then the path must then take an inter-gadget edge from v_{out} to some w_{in}, forcing the next gadget to also be traversed from the 'in' node to the 'out' node. By induction, all gadgets are traversed in this manner. (We can symmetrically argue for a gadget being entered at v_{out} and exited at v_{in}).

   Now suppose p is of the form

   $$v_{1_{in}},\ v_1,\ v_{1_{out}},\ v_{2_{in}},\ v_2,\ v_{2_{out}},\ \ldots,\ v_{n_{in}},\ v_n,\ v_{n_{out}}$$

then each inter-gadget edge between $v_{i_{out}}$ and $v_{(i+1)_{in}}$ corresponds to a directed edge $v_i \to v_{i+1}$ in G, so $v_1 \to v_2 \to \cdots \to v_n$ is a Hamiltonian path in G. If p is of the form

$$v_{1_{out}},\ v_1,\ v_{1_{in}},\ v_{2_{out}},\ v_2,\ v_{2_{in}},\ \ldots,\ v_{n_{out}},\ v_n,\ v_{n_{in}}$$

then inter-gadget edge between $v_{i_{in}}$ and $v_{(i+1)_{out}}$ corresponds to a directed edge $v_i \leftarrow v_{i+1}$ in G, so $v_1 \leftarrow v_2 \leftarrow \cdots \leftarrow v_n$ is a Hamiltonian path in G.

Thus, G has a Hamiltonian path if and only if H has a Hamiltonian path. (And this construction takes polynomial time because we are only adding three vertices per vertex.)